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To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Experiment No. 07

Aim: - Write a program to demonstrate
- use of implementing interfaces.
- use of extending interfaces.

Software Requirement: - Java Development Kit (JDK), Integrated Development Environment (IDE) (NetBeans), Java Runtime Environment (JRE).

Theory: -

a) Implementing Interface:

In java, an interface is implemented by a class. The class that implements an interface must provide code for all the methods defined in the interface, otherwise, it must be defined as an abstract class.

The class uses a keyword implements to implement an interface. A class can implement any number of interfaces. When a class wants to implement more than one interface, we use the implements keyword is followed by a comma-separated list of the interfaces implemented by the class.

An **Interface in Java** programming language is defined as an abstract type used to specify the behavior of a class. An interface in Java is a blueprint of a behavior. A Java interface contains static constants and abstract methods.

- The interface in Java is *a* mechanism to achieve abstraction.
- By default, variables in an interface are public, static, and final.
- It is used to achieve abstraction and multiple inheritances in Java.
- It is also used to achieve loose coupling.
- In other words, interfaces primarily define methods that other classes must implement.

The following is the syntax for defining a class that implements an interface.

Syntax

```
class className implements InterfaceName{
```

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```
    ...
    boby-of-the-class
    ...
}
```

b) Extending Interfaces:

- In java, an interface can extend another interface. When an interface wants to extend another interface, it uses the keyword extends.
- The interface that extends another interface has its own members and all the members defined in its parent interface too.
- The class which implements a child interface needs to provide code for the methods defined in both child and parent interfaces, otherwise, it needs to be defined as abstract class.
- One interface can inherit another by the use of keyword extends. When a class implements an interface that inherits another interface, it must provide an implementation for all methods required by the interface inheritance chain.

Syntax

```
interface ParentInterface {
    void method1();
}

interface ChildInterface extends ParentInterface {
    void method2();
}
```

Advantages and Disadvantages of Implementing and Extending Interfaces in Java

a) Implementing Interfaces

<u>Advantages</u>

- **Multiple Inheritance Support** – Since Java doesn't support multiple inheritance with classes, interfaces allow a class to inherit behavior from multiple sources.

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- **Decoupling & Flexibility** – Interfaces help in designing loosely coupled systems by defining contracts that different classes can follow.
- **Better Testability** – Makes unit testing easier by allowing mock implementations.
- **Encapsulation of Behavior** – Defines a clear contract for implementing classes without exposing implementation details.
- **Encourages Code Reusability** – Multiple classes can share common behavior without sharing a parent class.

Disadvantages

- **More Boilerplate Code** – Every implementing class must provide implementations for all interface methods, leading to redundant code.
- **No State Management** – Interfaces can't hold state (fields), meaning you must use classes for storing data.
- **Backward Compatibility Issues** – Adding new methods to an existing interface breaks all implementing classes unless default methods are used.
- **Indirect Access to Implementation** – You can't directly instantiate an interface, only reference it, which may add complexity.

b) Extending Interfaces

Advantages

- **Enhanced Modularity** – Breaks down complex behaviors into multiple smaller, manageable interfaces.
- **Code Reusability** – Avoids redundant declarations in multiple interfaces.
- **Supports Multiple Inheritance of Type** – A new interface can inherit from multiple interfaces, combining their behaviors.
- **Improves API Evolution** – New interfaces can be built on top of existing ones without modifying the old ones.

Disadvantages

- **Complexity in Hierarchy** – Extending too many interfaces can make it hard to track relationships.
- **Diamond Problem (Method Conflicts)** – If multiple interfaces define the same default method, conflicts must be resolved manually.
- **Potential Overhead** – If too many interfaces extend each other, it can lead to excessive dependencies and maintenance issues.

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Program and Output:

a) Implementing Interface:

```
interface Human {
        void learn(String str);
        void work();

        int duration = 10;

}

class Programmer implements Human{
        public void learn(String str) {
                System.out.println("Learn using " + str);
        }
        public void work() {
                System.out.println("Develop applications");
        }
}

public class HumanTest {

        public static void main(String[] args) {
                Programmer trainee = new Programmer();
                trainee.learn("coding");
                trainee.work();
        }
}
```

Output:

```
Learn using coding
Develop applications
```

b) Extending Interfaces:

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```
interface ParentInterface{
	void parentMethod();
}

interface ChildInterface extends ParentInterface{
	void childMethod();
}

class ImplementingClass implements ChildInterface{

	public void childMethod() {
		System.out.println("Child Interface method!!");
	}

	public void parentMethod() {
		System.out.println("Parent Interface mehtod!");
	}
}

public class ExtendingAnInterface {

	public static void main(String[] args) {

		ImplementingClass obj = new ImplementingClass();

		obj.childMethod();
		obj.parentMethod();

	}

}
```

Output:
Child Interface method!!
Parent Interface mehtod!

Conclusion: Hence, in this practical, we learned and executed the use of implementing interfaces and use of extending interfaces in Java Successfully.